

Sizing a Rooftop Solar Array

Finding the area of an oblique triangle from survey bearings

First Member Second Member Third Member

Group Activity — Trigonometry

August 16, 2026

Outline

- 1 The problem
- 2 From bearings to an angle
- 3 Finding the area
- 4 Discussion

The brief

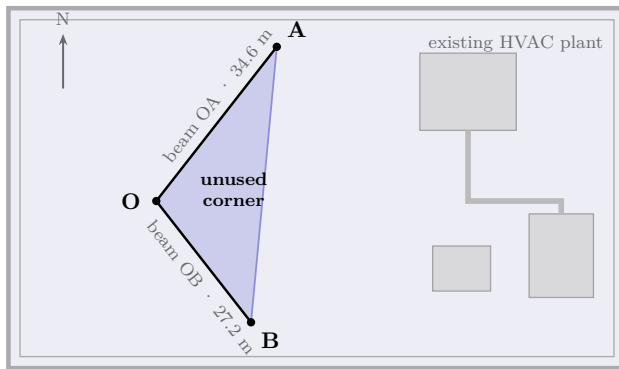
An engineering firm is scoping a small solar array in an unused triangular corner of a warehouse rooftop, boxed in by two existing structural beams. They need **the enclosed area** — that is what sets how many panels the site can hold.

The catch

The surveyor never recorded an angle. The field notes contain **two bearings**, which is how rooftop and land surveys are actually reported.

So the job splits in two: recover the angle, then find the area.

The site



Plan view · warehouse rooftop bounded by two structural beams

What the surveyor actually recorded

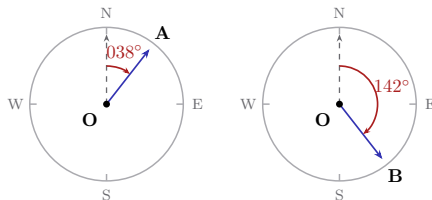
Field notes — station O

Beam OA 34.6 m, bearing 038°

Beam OB 27.2 m, bearing 142°

Angle at O *not recorded*

A total station measures **distance** and **direction**, and direction is logged as a bearing — nobody stands in the corner with a protractor. **So the included angle has to be derived.**



A bearing is always measured *clockwise from north* — same origin, same sense, every time.

From two bearings to one angle

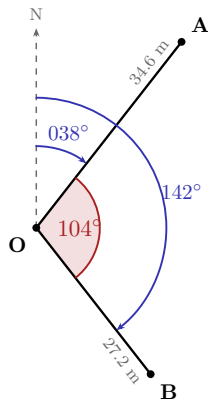
Two conditions make this a plain subtraction:

- both bearings share the **same reference point, O**;
- both are measured **clockwise from north**.

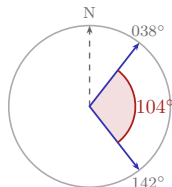
Included angle

$$\angle AOB = 142^\circ - 38^\circ = 104^\circ$$

No trigonometry yet — this step is bookkeeping about how the directions were recorded.

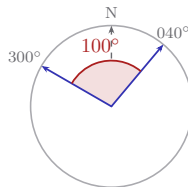


The rule in general



$$142 - 38 = 104 < 180$$

take it directly



$$300 - 40 = 260 > 180$$

use $360 - 260 = 100^\circ$

Subtract the bearings; if the result exceeds 180° you have measured the *reflex* side, so take 360° minus it instead. Here $104^\circ < 180^\circ$, so no adjustment is needed.

Now it is a standard SAS triangle

Known

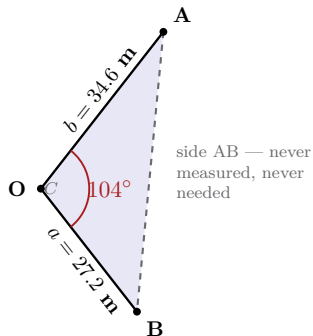
$$a = OB = 27.2 \text{ m}$$

$$b = OA = 34.6 \text{ m}$$

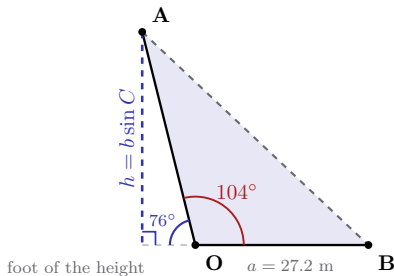
$$C = 104^\circ \text{ (derived)}$$

The angle sits **between** the two known sides — exactly the SAS pattern.

C is obtuse, so the triangle is genuinely oblique: no right angle to fall back on, and no perpendicular height in the field notes.



Why $\frac{1}{2}ab \sin C$ is the right tool



$$\begin{aligned}\text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} a (b \sin C)\end{aligned}$$

C is obtuse, so the foot lands outside the triangle — and $\sin 104^\circ = \sin 76^\circ$ still holds.

The familiar $\frac{1}{2} \times \text{base} \times \text{height}$ never went away. $\sin C$ simply supplies the height from the side and angle we do have.

The calculation

$$\begin{aligned}\text{Area} &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 34.6 \times 27.2 \times \sin 104^\circ \\ &= 470.56 \times \sin 104^\circ \\ &= 470.56 \times 0.9703 \\ &\approx 456.58 \text{ m}^2\end{aligned}$$

Watch for

Calculator in **degrees**, not radians.

$\sin 104^\circ = \sin 76^\circ = 0.9703$ — an obtuse angle gives a perfectly ordinary positive sine.

Round only at the **final step**.

Complete solution

Quantity	Value
Side OA (b)	34.6 m
Side OB (a)	27.2 m
Bearing to A	038°
Bearing to B	142°
Included angle C	$142^\circ - 38^\circ = 104^\circ$
$\sin 104^\circ$	≈ 0.9703
$\frac{1}{2} \times \text{OA} \times \text{OB}$	470.56
Area	$\approx 456.58 \text{ m}^2$

The answer

Area of the triangular rooftop section

$$\text{Area} = \frac{1}{2} \times 34.6 \times 27.2 \times \sin 104^\circ \approx 456.58 \text{ m}^2$$

From two beam lengths and two bearings — no angle ever measured.

A note on rounding

$\sin 104^\circ = 0.970296\dots$, so

$$470.56 \times 0.970296 = 456.582\dots$$

A rounded 0.9709 would give
 456.87 m^2 — about 0.3 m^2 adrift,
roughly **one panel's worth** of area.

Small, but the kind of error that
compounds: carry the full value
through and round once, at the end.

Sensitivity of the answer

$$\sin C = 0.9703 \quad 456.58 \text{ m}^2$$

$$\sin C = 0.9709 \quad 456.87 \text{ m}^2$$

$$\text{difference: } 0.29 \text{ m}^2$$

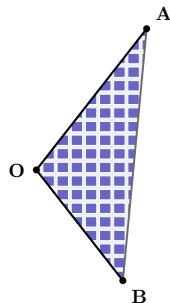
Optional: what could that area produce?

Utility-scale arrays are often quoted at roughly 170 W of installed capacity per square metre:

$$456.58 \times 170 \approx 77\,619 \text{ W} \approx 77.6 \text{ kW}$$

Say this out loud

A rough, order-of-magnitude estimate — not an engineering calculation. It ignores tilt, row spacing, shading and inverter losses, all of which reduce the real figure.



schematic only — tilt, row spacing
and shading are not modelled

What to take away

1. The angle was earned, not given

Bearings share a reference direction and a rotational sense, so their difference *is* the angle between the lines. Real survey data arrives this way; converting it is part of the problem.

2. It is still just base $\times$ height

$\frac{1}{2}ab \sin C$ is not a new formula to memorise. $\sin C$ stands in for a perpendicular height that was never measured — which is why SAS is enough.

From *two distances and two directions* to **456.58 m²** — and a panel count.

Questions?